
Homework 1
MAT 3510
Due Date: Sept. 15, 2026

Instruction: Your homework should follow the requirements as in the syllabus.

Problem 1 [10 points]

In statistics, one define the variance of sample values $x_1, \dots, x_n$ by

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Write a MATLAB function

`function s2 = samplevar (x)`

that takes as input a vector x of any length and returns s^2 as calculated by the formula. You should test your function on some data and compare it to MATLAB's built in `var`.

Problem 2 [10 points]

Write a MATLAB function

`function r = polyadd (p,q)`

that returns the coefficient vector r for the sum of two polynomials p and q , specified as vectors of coefficients in decreasing degree. You should *not* assume that p and q are given by vectors of the same length.